

Problem Set 1

In this problem set, we look at demographic data from Sweden, Kenya, and the *world*, to compare birth and death rates over a 50 year period.

Crude Birth Rates

First, let's compute crude death rates for each country and period.

```
# read in the data
kenya <- read.csv('INTRO/Kenya.csv')
sweden <- read.csv('INTRO/Sweden.csv')
world <- read.csv('INTRO/World.csv')

# create a function to compute crude birth rates
calculate_cbr <- function(dataset, period){

  # total person-years column
  dataset$py <- dataset$py.women + dataset$py.men

  # compute total person-years and births in the given period
  py_total_period <- sum(dataset$py[dataset$period == period])
  births_total_period <- sum(dataset$births[dataset$period == period])

  # return crude birth rate
  return(births_total_period / py_total_period)
}

calculate_cbr(kenya, '1950-1955')
```

```
[1] 0.0520949
```

```
calculate_cbr(kenya, '2005-2010')
```

```
[1] 0.03851507
```

```
calculate_cbr(sweden, '1950-1955')
```

```
[1] 0.01539614
```

```
calculate_cbr(sweden, '2005-2010')
```

```
[1] 0.01192554
```

```
calculate_cbr(world, '1950-1955')
```

```
[1] 0.03732863
```

```
calculate_cbr(world, '2005-2010')
```

```
[1] 0.02021593
```

Age-Specific Fertility Rates

Next, let's compute age-specific fertility rates (ASFR) for women aged 15-49.

```
asfr <- function(dataset, period){  
  # just keep the women 15-50 in the period of interest  
  dataset <- dataset[dataset$period == period & dataset$age %in% c('15-19', '20-24', '25-29',  
    asfr <- dataset$births / dataset$py.women  
    names(asfr) <- dataset$age  
    return(asfr)  
  }  
  asfr(dataset = kenya, period = '1950-1955')
```

15-19	20-24	25-29	30-34	35-39	40-44	45-49
0.16884585	0.35596942	0.34657814	0.28946367	0.20644016	0.11193267	0.03905205

```
asfr(dataset = kenya, period = '2005-2010')
```

15-19	20-24	25-29	30-34	35-39	40-44	45-49
0.10057087	0.23583536	0.23294721	0.18087964	0.13126805	0.05626214	0.03815044

```
asfr(dataset = sweden, period = '1950-1955')
```

15-19	20-24	25-29	30-34	35-39	40-44
0.038908952	0.127710883	0.125243665	0.087364159	0.048603771	0.016210186
45-49					
0.001341829					

```
asfr(dataset = sweden, period = '2005-2010')
```

15-19	20-24	25-29	30-34	35-39	40-44
0.0059709097	0.0507320271	0.1162085625	0.1322744621	0.0625923991	0.0121600765
45-49					
0.0006143942					

that's interesting. Sweden birth rates declined in every age group *except* women in their

```
asfr(dataset = world, period = '1950-1955')
```

15-19	20-24	25-29	30-34	35-39	40-44	45-49
0.09029521	0.23763370	0.25245229	0.20416410	0.13810534	0.06360832	0.01519064

```
asfr(dataset = world, period = '2005-2010')
```

15-19	20-24	25-29	30-34	35-39	40-44
0.048489719	0.151971307	0.146980966	0.093813813	0.046689639	0.016268995
45-49					
0.004510245					

Total Fertility Rate

Now let's use the ASFRs to compute the total fertility rate (TFR) for each country and period.

```
tfr <- function(dataset, period){  
  asfr <- asfr(dataset, period)  
  # total fertility rate is each asfr times 5, added together  
  sum(asfr * 5)  
}  
  
tfr(kenya, '1950-1955')
```

```
[1] 7.59141
```

```
tfr(kenya, '2005-2010')
```

```
[1] 4.879568
```

```
tfr(sweden, '1950-1955')
```

```
[1] 2.226917
```

```
tfr(sweden, '2005-2010')
```

```
[1] 1.902764
```

```
tfr(world, '1950-1955')
```

```
[1] 5.007248
```

```
tfr(world, '2005-2010')
```

```
[1] 2.543623
```

Crude Death Rate

Next, we'll compute crude death rates (CDR) for each country and period.

```
calculate_cdr <- function(dataset, period){  
  
  # total person-years column  
  dataset$py <- dataset$py.women + dataset$py.men  
  
  # compute total person-years and deaths in the given period  
  py_total_period <- sum(dataset$py[dataset$period == period])  
  deaths_total_period <- sum(dataset$deaths[dataset$period == period])  
  
  # return crude death rate  
  return(deaths_total_period / py_total_period)  
}  
  
calculate_cdr(kenya, '1950-1955')
```

```
[1] 0.02396254
```

```
calculate_cdr(kenya, '2005-2010')
```

```
[1] 0.01038914
```

```
calculate_cdr(sweden, '1950-1955')
```

```
[1] 0.009844842
```

```
calculate_cdr(sweden, '2005-2010')
```

```
[1] 0.009968455
```

```
# interesting note here. crude death rates appear to be equal in Kenya and Sweden in the 2000s
```

```
calculate_cdr(world, '1950-1955')
```

```
[1] 0.01931893
```

```
calculate_cdr(world, '2005-2010')
```

```
[1] 0.008166083
```

Age-Specific Death Rate

Now let's compute age-specific death rates (ASDR) for each country and period.

```
asdr <- function(dataset, period){  
  # just keep the period of interest  
  dataset <- dataset[dataset$period == period,]  
  
  asdr <- dataset$deaths / (dataset$py.women + dataset$py.men)  
  names(asdr) <- dataset$age  
  
  return(asdr)  
}  
  
asdr(kenya, '1950-1955')
```

0-4	5-9	10-14	15-19	20-24	25-29
0.066826532	0.009321789	0.005972093	0.005869582	0.007651103	0.008838750
30-34	35-39	40-44	45-49	50-54	55-59
0.009677594	0.010986891	0.012633744	0.014760408	0.018260395	0.024433007
60-69	70-79	80+			
0.041996801	0.093683927	0.200016381			

```
asdr(kenya, '2005-2010')
```

0-4	5-9	10-14	15-19	20-24	25-29
0.020920755	0.002911301	0.002918895	0.002942986	0.003885368	0.006558131
30-34	35-39	40-44	45-49	50-54	55-59
0.010603913	0.013881062	0.013474598	0.011288057	0.011152339	0.013898334
60-69	70-79	80+			
0.025395531	0.061261551	0.158620510			

```
asdr(sweden, '1950-1955')
```

0-4	5-9	10-14	15-19	20-24	25-29
0.0047456697	0.0004320537	0.0004896406	0.0007431865	0.0010177339	0.0011140910
30-34	35-39	40-44	45-49	50-54	55-59
0.0013343851	0.0017429491	0.0025095541	0.0039668755	0.0063486410	0.0101672774
60-69	70-79	80+			
0.0214156644	0.0599823093	0.1678170255			

```
asdr(sweden, '2005-2010')
```

0-4	5-9	10-14	15-19	20-24	25-29
6.790712e-04	8.138094e-05	1.135496e-04	2.687775e-04	4.697344e-04	4.941440e-04
30-34	35-39	40-44	45-49	50-54	55-59
5.057066e-04	6.689578e-04	1.039256e-03	1.769621e-03	2.988715e-03	4.709913e-03
60-69	70-79	80+			
9.828772e-03	2.803963e-02	1.098892e-01			

```
asdr(world, '1950-1955')
```

0-4	5-9	10-14	15-19	20-24	25-29
0.054589755	0.005600412	0.004261869	0.004752908	0.005891020	0.006325420
30-34	35-39	40-44	45-49	50-54	55-59
0.007132501	0.008534487	0.010572557	0.013459846	0.017335769	0.024265320
60-69	70-79	80+			
0.042262017	0.086910343	0.184364978			

```
asdr(world, '2005-2010')
```

0-4	5-9	10-14	15-19	20-24	25-29
0.012802492	0.001256903	0.001079067	0.001302818	0.001832602	0.002278500
30-34	35-39	40-44	45-49	50-54	55-59
0.002623982	0.003031563	0.003753402	0.005085583	0.007126588	0.010477192
60-69	70-79	80+			
0.020235894	0.047457519	0.120679385			

Counterfactual CDR

Finally, what would Kenya's Crude Death Rate be if it had the age-distribution of Sweden?

```
asdr(kenya, '2005-2010')
```

0-4	5-9	10-14	15-19	20-24	25-29
0.020920755	0.002911301	0.002918895	0.002942986	0.003885368	0.006558131
30-34	35-39	40-44	45-49	50-54	55-59
0.010603913	0.013881062	0.013474598	0.011288057	0.011152339	0.013898334
60-69	70-79	80+			
0.025395531	0.061261551	0.158620510			

```
sweden$py <- sweden$py.men + sweden$py.women
sweden_population <- sum(sweden$py[sweden$period=='2005-2010'])
sweden_period <- sweden[sweden$period == '2005-2010',]
# the proportion column will now tell me the percent of Swedes in each age bucket during the
sweden_period$proportion <- sweden_period$py / sweden_population

# counterfactual CDR is sum(asdr x sweden's population proportions)
sum(asdr(kenya, '2005-2010') * sweden_period$proportion)
```

```
[1] 0.02321646
```